

Screen Recapture Detection using Handcrafted Features and Random Forest

1. Objective

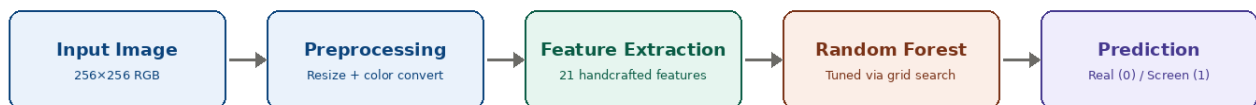
This project addresses a specific fraud pattern: distinguishing a genuine photograph from a photo taken of a screen or printout displaying an image (a “recapture”). The goal was a solution that is accurate, but also small, fast, and cheap enough to eventually run on a phone , not just a model trained for accuracy alone.

2. Dataset

A self-collected dataset of 120 images was used — 60 real photographs and 60 screen recaptures (balanced to avoid class bias). Real photos covered varied indoor objects, distances, and lighting; screen photos were created by displaying the same kinds of images on a laptop or phone and re-photographing them at different angles, distances, and glare levels, so the model learns recapture artifacts rather than scene content.

3. Approach

Given the dataset size, a deep CNN risked overfitting to this specific phone and these screens rather than learning a generalizable signal. Instead, 21 handcrafted features were extracted per image, grouped into five categories: sharpness/texture (blur score, edge density, RMS contrast, Tenengrad, LBP mean, LBP variance), frequency content (FFT high frequency, FFT low frequency, pixel grid score — for moiré and sub-pixel grid patterns), color/tone (RGB means and standard deviations, saturation mean, brightness), gradients (Sobel X, Sobel Y), and statistics (entropy, noise estimate). These feed a Random Forest classifier, tuned via grid search with 5-fold cross-validation over tree count, depth, split criteria, and class balancing — chosen for strong performance on small tabular feature sets and fully traceable, interpretable decisions.



4. Results

Metric	Result
Dataset	120 images (60 real, 60 screen)
Features	21 handcrafted features
Model	Random Forest (grid-search tuned)
Test accuracy	87.5%
Average latency	67.01 ms / image (CPU)
Deployment cost	≈ \$0 (on-device inference)

5. Limitations

The main limitation is dataset size: 120 self-collected images may not capture every device, lighting setup, or display type in the held-out set, so real-world accuracy will likely differ from the reported 87.5%. All three

misclassifications were screen photos predicted as real, with zero real photos misclassified as screen — the model is more conservative than it should be on screen photos with weak glare or moiré, at a flat angle or from further away.

6. Future Work

Future work include a larger, more device-diverse dataset; testing a lightweight CNN such as MobileNetV2 as a complementary signal and adding screen-specific texture descriptors (e.g. GLCM) to catch the weak-glare cases the current features tend to miss.